Conservation of rare populations

Learning objectives

1. Understand the causes and consequences of small or rare populations
2. Know the key terms and process of population viability analysis

What causes populations to be small or rare?

- natural causes:
  - endemism/habitat specialization; Maguire's primrose in Logan Canyon *limestone cliffs at certain aspects and elevations*
  - dioecious (separate male and female plants); ginkgo trees
  - mutualist specialization; fig wasps
  - dispersal limitation
- anthropogenic causes:
  - habitat fragmentation
  - overexploitation; ornamental plants
  - invasive species
  - climate change

Population dynamics in rare populations

Demographic stochasticity

- variation due to chance events affecting individuals
- increases extinction risk in small populations, ignore in large populations
- reproductive individuals randomly have few offspring for a few years -> extinction risk

plot two lines, large and small population with fluctuating populations, small pop goes to zero

- extinction risk increases when combined with environmental stochasticity *in models*
- bad year -> lower population size -> demographic stochasticity extinction risk

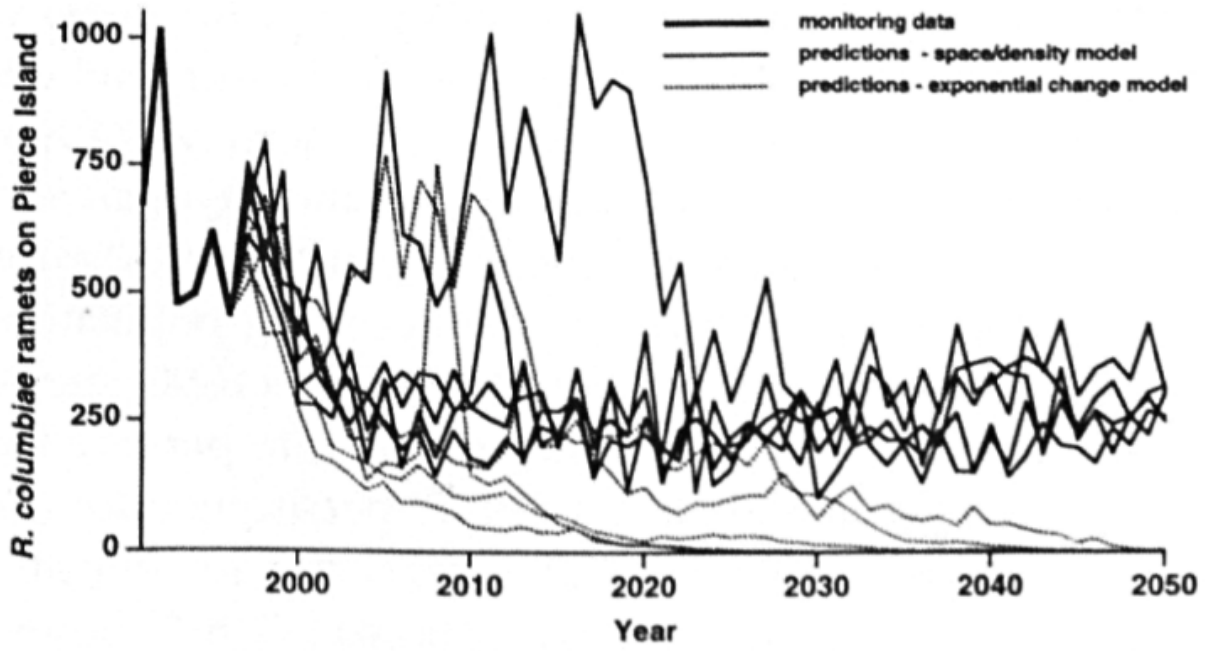
Inbreeding depression

- reduction in fitness when closely related individuals reproduce
- levels of relatedness:
  - self-fertilization
  - outcrossing within a small population
  - outcrossing between small populations
- small populations -> low genetic diversity -> more harmful alleles
- examples:
  - stunted growth
  - smaller inflorescences
  - fewer seeds

Modeling

Population viability analysis

- terms
  - persistence time: probability of persistence for given period of time
  - extinction probability: proportion of simulations that end in extinction
  - time to extinction: median or mean number of years to extinction
- population model + simulations projecting past trends forward
  - multiple population models
  - density dependence, env. heterogeneity, etc.
  - simulations: include stochasticity
  - can add management scenarios



Managing rare populations

- depends on the cause of population decline
  - example: habitat protection for fragmentation
- assisted migration: moving plants to areas with suitable climate
  - survive climate change *can't otherwise disperse to sutiable habitat*
  - risks: aggressive genotypes, outbreeding depression (reduced fitness when too genetically distant)
  - example: whitebark pine *threatened by pathogens, pests, and climate change; transplants in northern edge in Canada*